Python Practice: Mixed Data Structures

Part 5 of 5 | 10 questions | Beginner to Expert

RULES: No for/while loops, no comprehensions, no recursion, no map/filter/reduce, and no hardcoded answers. Use indexing, slicing, operators, built-in functions, and data structure methods.

LEVEL: ADVANCED

Q41. Student Performance System

```
students = [
    {"id": 1, "name": "Ali", "marks": (85, 90, 88)},
    {"id": 2, "name": "Sara", "marks": (92, 95, 91)},
    {"id": 3, "name": "Ahmed", "marks": (78, 84, 80)}
]
```

Calculate each average, top student, tuple of names, and set of unique marks.

Q42. Employee Skills

```
employees = {
    "Ali": {"Python", "SQL", "AWS"},
    "Sara": {"Python", "Databricks", "SQL"},
    "Ahmed": {"AWS", "Snowflake", "SQL"}
}
```

Find common skills, Ali-only skills, all team skills, and employees meeting Python + SQL requirements.

Q43. Sales Data

```
sales = [
    ("Laptop", 120000, 2),
    ("Mouse", 1500, 5),
    ("Keyboard", 3500, 3)
]
```

Calculate revenue, top-revenue product, name-to-revenue dictionary, and set of prices.

Q44. Customer Orders

```
customers = {
    "C101": {"name": "Ali", "orders": [1200, 2500, 1800]},
    "C102": {"name": "Sara", "orders": [3000, 4500]},
    "C103": {"name": "Ahmed", "orders": [1500, 2000, 1000]}
}
```

Calculate spending per customer, top customer, and tuple of totals in customer ID order.

Q45. Course Enrollment

```
courses = {
    "Python": {"Ali", "Sara", "Ahmed"},
    "SQL": {"Sara", "Ahmed", "Zain"},
    "AWS": {"Ali", "Ahmed", "Zain"}
}
```

Find students in exactly two, all three, and exactly one course; count unique students.

LEVEL: EXPERT

Q46. E-Commerce Transactions

```
transactions = [
    {"id": 101, "customer": "Ali", "items": [("Laptop", 120000, 1), ("Mouse", 1500, 2)]},
    {"id": 102, "customer": "Sara", "items": [("Keyboard", 3500, 2), ("Mouse", 1500, 1)]},
    {"id": 103, "customer": "Ali", "items": [("Monitor", 45000, 1)]}
]
```

Calculate total revenue, Ali spending, largest transaction, and unique product names.

Q47. Pipeline Monitoring

```
pipelines = {
    "pipeline_1": {"status": "success", "records": (1000, 1200, 1100)},
    "pipeline_2": {"status": "failed", "records": (800, 0, 0)},
    "pipeline_3": {"status": "success", "records": (1500, 1600, 1700)}
}
```

Find total processed records, failed pipelines, largest pipeline total, and average records per run for each.

Q48. Banking Transactions

```
accounts = {
    "A101": {"holder": "Ali", "transactions": [5000, -2000, 3000, -1000]},
    "A102": {"holder": "Sara", "transactions": [10000, -3000, -2000]},
    "A103": {"holder": "Ahmed", "transactions": [7000, -1500, 2500]}
}
```

Assuming zero starting balances, calculate final balances, highest balance, total withdrawals, and largest single deposit.

Q49. Product Recommendations

```
users = {
    "Ali": {"purchased": {"Laptop", "Mouse"}, "wishlist": {"Keyboard", "Monitor", "Mouse"}},
    "Sara": {"purchased": {"Keyboard", "Monitor"}, "wishlist": {"Laptop", "Mouse"}},
    "Ahmed": {"purchased": {"Laptop", "Keyboard"}, "wishlist": {"Monitor", "Headphones"}}
}
```

Find products bought by at least two users, Ali wishlist minus purchases, shared wishlist products, and all products.

Q50. Final Company Analysis

```
company = {
    "departments": {
        "Engineering": {
            "employees": [
                ("Ali", 120000, {"Python", "AWS", "SQL"}),
                ("Sara", 150000, {"Python", "Databricks", "SQL"})
            ],
            "budget": 500000
        },
        "Analytics": {
            "employees": [
                ("Ahmed", 100000, {"SQL", "Power BI"}),
                ("Zain", 130000, {"Python", "SQL", "Tableau"})
            ],
            "budget": 400000
        }
    }
}
```

1. Payroll per department. 2. Company payroll. 3. Highest-paid employee. 4. Skills shared by all. 5. Engineering-only skills. 6. Remaining budget per department. 7. Department with higher average salary. 8. Name-to-salary dictionary. 9.

All unique skills. 10. Is combined budget sufficient?